Amjad S Althabteh

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Summary

Built rendering engines from scratch. Optimized GPU pipelines for sub-millisecond frame budgets. Shipped diagnostics tooling trusted across 10K+ test runs. C++ systems engineer with deep expertise in C++17–C++23, OpenGL, and performance-critical design.

Education

B.S. in Computer Science — Wayne State University
Detroit, MI | GPA: 3.65/4.00

Experience

Graphics C++ Engineer **Immersalab Graphics** ***Detroit, MI*** **Sep 2025 - Dec 2025**

- Architected a C++20/OpenGL real-time rendering engine with GPU-driven pipelines, achieving ~25% improvement in stability
- Built internal UI tools and a JSON asset pipeline.
- Implemented data-driven optimizations to balance GPU workloads, reducing GPU idle time by **~30%** across test scenes
- Developed GPU profiling and frame-timing tools, cutting render bottleneck diagnosis time.

Indie Game Dev **Unreal C++ Developer** ***Ann Arbor, MI*** **Apr 2025 - Aug 2025**

- Developed **8+ gameplay systems** in C++20 and Unreal blueprints integrating animation state machines.
- Produced a 90-second real-time cinematic using MetaHuman, control rig, and sequencer.
- Built animation control systems supporting smooth multi-layer blending.

Software Development Intern **Rare Munchies** ***Ann Arbor, MI*** ***Sep 2023 - Apr 2024***

- Integrated UI components using TypeScript, improving page transition speed by **~20%** and interaction responsiveness
- Identified and resolved 8+ cross-browser compatibility issues.
- Optimized client-side rendering logic and state management, reducing unexpected UI re-renders by **~35%**

Undergraduate Researcher | Orbital Collision Modeling
Wayne State University — Detroit, MI | Feb 2025 – Dec 2025

- Modeled satellite trajectories using numerical integration & N-body gravitational dynamics
- Simulated orbital collision probabilities across trajectory samples using spatial partitioning, reducing compute time by ~40%

Projects

Glint 3D | C++ Rendering Engine *glint-rendering*

- Engineered a **C++17 OpenGL rendering engine** with a custom scene graph.
- IImplemented frustum culling and geometry instancing via glDrawElementsInstanced.
- Designed a fixed-size memory pool allocator using placement new and freelist recycling, reducing heap allocations by ~60%

3D Space Simulation Engine | C++ | OpenGL *space-simulations*

- Developed a C++17/OpenGL engine supporting dynamic lighting across 10K+ simulated entities with stable frame delivery
- Designed 4 modular engine systems (rendering, physics, math, entity) with clean separation of concerns
- Implemented a custom math library (Vec3, Mat4) and camera system supporting full 3D projection/view pipeline.

HFT OrderBook Engine | C++20

- Implemented a FIFO matching engine in C++20 processing **~2.1M orders/sec** w/ sub-microsecond latency.
- Used alignas(64) cache-aligned bid/ask structs to eliminate false sharing and reduce memory latency.

Traceforge | C++ Runtime Stack Trace & Diagnostics Engine

- Built a C++23 diagnostics engine using std::stacktrace across 5+ build configs, cutting debug triage time by ~60%
- Implemented custom operator new/delete overloads for memory tracking across 10K+ test runs
- Automated regression snapshot diffing using std::filesystem across build artifacts.

Skills

Programming: C++ (C++11–C++23), C, C#, Python, TypeScript, Java, SQL
Graphics, Engines & Tools: OpenGL 4.x, GLSL, Unreal Engine 4–5, Unity, Godot, CMake, VS Code, Git
Systems & Performance: RAII, custom allocators, memory pools, lock-free data structures, cache optimization, profiling (Valgrind, perf)